Sahaj Bhadja

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Summary

Software Engineer with 2+ years building and testing reliable **cross-platform software**, automation, and data systems using **C++**, Python, Windows, OOP, CI/CD, and Agile development practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built data integrations across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD-backed releases achieving 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Implemented a production backend on AWS serving 150+ instructors and 10K+ daily grading requests, using CI/CD-driven workflows to reduce feedback turnaround 60% while maintaining stable API performance.
- Designed concurrent batch processing and secure PDF, DOCX, and ZIP ingestion through Python REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery mechanisms.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Engineered cross-platform data-sharing software with microservices and CI/CD pipelines for unit and integration testing across Windows, Linux, and macOS while collaborating with Microsoft and GitHub stakeholders in Agile sprints.
- Automated nightly builds, executable installation, integration tests, unit tests, and CSV reports through Jenkins, validating reliable cross-platform synchronization on virtual machines while reducing manual entry by 35%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight object-oriented microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Automated Docker-based deployments and software validation through Jenkins CI/CD, achieving 95%+ successful build checks while reducing deployment time 50% across controlled staging and production software release environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Flask APIs, payment webhooks, pricing, and coupon workflows while synchronizing Firebase booking events and processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Languages & Systems: C++, Python, Java, JavaScript, TypeScript, SQL, OOP, Data Structures, Algorithms, Windows, Linux, macOS
Testing & Development: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Validation, Requirements Analysis, Code Reviews, Software Documentation, Agile/Scrum
Tools & Delivery: Git, GitHub, Jenkins, GitHub Actions, CI/CD, Docker, Virtual Machines, REST APIs, Microservices, Postman
Data & Cloud: AWS, Azure, Azure SQL, PostgreSQL, SQL Server, MySQL, MongoDB, Firebase, S3, AWS Glue, FastAPI, Flask, React

Projects

CPU-scheduling Simulator Live | *C++, React, Operating Systems, Algorithm Verification*

- Implemented six C++ scheduling algorithms with edge-case handling, React controls, Gantt visualization, and dynamic performance metrics, systematically testing functional correctness across 1K+ preemptive and non-preemptive process combinations.

Chat2SQL Code | *Java, TypeScript, Docker, Windows, Linux, macOS*

- Worked across Java and TypeScript modules using Docker, Maven, Node.js, and Yarn to support tested cross-platform database workflows for 10+ systems on Windows, macOS, and Linux environments.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Sentry*

- Implemented consent-based patient-history access, healthcare workflows, appointment alerts, and Sentry monitoring through scalable TypeScript services supporting 100+ daily clinical operations while reducing hospital administrative workload by 30%.